# Can Polymarket Become a More Accurate "Survey"? A Replacement for Traditional Surveys That Are Easy to Game?

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**Abstract**

Traditional surveys often face incentive problems: respondents may answer carelessly, answer just to finish quickly (especially in paid surveys), or answer according to social desirability without any direct consequence. Prediction markets such as Polymarket offer an alternative mechanism: participants stake their own money, so incorrect predictions have real costs. Academic literature suggests that prediction markets are often quite accurate and, in many contexts, can outperform polling—especially at longer time horizons.

This paper (i) explains how Polymarket works (binary contracts, USDC collateral on Polygon, a CLOB order book mechanism, and oracle-based resolution), (ii) summarizes empirical evidence from the broader prediction-market literature, and (iii) incorporates quantitative findings from Polymarket data studies. An important note: when a market's resolution criteria are ambiguous, the price is often better interpreted as the probability that the market will be *resolved* as Yes under the stated rules, rather than the probability that the underlying real-world event "truly" happened [20, 19, 16]. For example, Chen et al. (2024) analyze more than 15,000 addresses across 4,500 events and 8,500 markets [21], while Saguillo et al. (2025) report an estimated realized arbitrage profit of about $40 million [22].

All factual numbers/claims in this paper are stated as *results according to the cited sources*, not as independent claims by the author.

However, prediction markets are not "pure truth": participant bias, liquidity, contract design and resolution definitions, and manipulation risks (price impact, wash trading, and oracle-mechanism attacks) can limit how prices should be interpreted as probabilities.

## 1 Introduction and Research Questions

### 1.1 The core problem in surveys: incentives and answer quality

Surveys typically measure opinions or expectations through direct questions. However, survey answer quality often deteriorates when:

- respondents have **no consequences** if their answers are wrong or inconsistent,
- respondents are **optimizing for rewards** ("paid to fill in"), so the main incentive is to finish quickly,

- **social desirability bias** (answers are made to look "proper"), or **framing effects** from question wording.

In contrast, in prediction markets, "respondents" stake personal funds: wrong predictions produce real losses. This is the core argument for why Polymarket is often considered a more stringent "survey."

### 1.2 Research questions

This paper addresses three questions:

1. **How does Polymarket work mechanically**? (contracts, pricing, and resolution)
2. **Why can prediction markets be more accurate than surveys**? (incentives + evidence)
3. **How can whale manipulation happen, and what is the evidence**? (mechanisms + case studies)

### 1.3 Contributions and scope

The main contribution of this paper is to connect three layers of argument in one narrative: (i) an incentive argument (skin in the game) explaining *why* prediction markets can be more disciplined than surveys, (ii) a mechanism argument (how contracts and resolution work) explaining *what* a market "price" is actually predicting, and (iii) a risk argument (whales/wash trading/oracles) explaining *when* prices can be misleading.

### 1.4 Related work: has this topic been discussed before?

Yes. In general, the theme "prediction markets vs polling" has long been discussed in economics and forecasting. Wolfers and Zitzewitz summarize early evidence that prediction markets are often informative and can produce good forecasts across many domains [1, 2].

Specifically for **Polymarket**, several on-chain data studies and working papers have appeared and provide quantitative datasets that can be cited directly. For example, Chen et al. (2024) construct a *Political Betting Leaning Score* (PBLS) and analyze user behavior by combining data from more than 15,000 addresses, 4,500 events, and 8,500 markets [21].

Saguillo et al. (2025) study arbitrage due to mispricing across assets that should complement each other (the aggregate price should ideally equal $1) and report an estimated realized profit of about $40 million extracted from Polymarket arbitrage opportunities [22].

Capponi et al. (2025) use historical resolved Polymarket markets to test a "semantic trading" approach. They report relationship-prediction accuracy of roughly 60–70% and that relation-based trading strategies achieve about 20% *average returns* over a horizon of about one week [23].

On the public debate side, discussions about Polymarket as a "probability" signal for politics were prominent in 2024 media (e.g., Vox discusses how election betting odds are treated like "polls" and the risks of interpretation) [9].

## 2 How Polymarket Works (Mechanisms, Prices, and Resolution)

### 2.1 Contracts, "shares," collateral, and reading prices

Polymarket generally offers binary markets ("Yes/No"): a "Yes" position pays out if the event happens, and a "No" position pays out if it does not. In practice, users treat these positions like tradable *shares* that can be bought, sold, and traded before market end.

#### 2.1.1 Collateral: USDC on Polygon (why this matters for the "data" we read)

Unlike surveys that merely collect answers, Polymarket requires **collateral**. Polymarket documentation explains that transactions and deposits run on the Polygon network using USDC (and, in some bridge integrations, USDC.e) as trading collateral [7, 8]. This matters for this paper because Polymarket "probabilities" emerge from economic decisions made under that collateral system.

#### 2.1.2 Positions and payouts (a simple financial intuition)

For binary markets, the payout intuition is: you pay a price to obtain a claim that pays in the future if a given outcome occurs. If you buy "Yes" and the final outcome is Yes, you receive full value; if the outcome is No, your claim becomes worthless. Because positions can be traded before resolution, you can profit or lose without waiting for the end (e.g., selling "Yes" after the price rises).

#### 2.1.3 Price as probability (and why it is not always 1:1)

Conceptually, the "Yes" price is often read as the market-implied probability. However, interpreting "price = probability" is not always exact due to risk preferences, position constraints, transaction costs, and contract design [3]. For the purpose of a "survey," this means we should not treat prices as objective probabilities without checking calibration.

### 2.2 Trading and why prices move

Prices move because people buy/sell (or provide liquidity) when they believe the current price is too low/high relative to information they have. Three sources of price movement are important to distinguish:

1. **New information** (news, data releases, credible rumors).
2. **Rebalancing/hedging** by large traders.
3. **Noise/manipulation** in low-liquidity markets.

#### 2.2.1 Matching mechanism: a hybrid CLOB (Central Limit Order Book)

Technically, Polymarket uses a hybrid-decentralized CLOB: orders are created as user-signed EIP-712 messages, then matched/ordered off-chain by an operator, while settlement occurs on-chain in a non-custodial manner via the Exchange contract [17].

Implications for reading Polymarket as a "survey":

- You can observe **depth** (bid/ask), spreads, and order imbalance as signals—not only last price.
- Orders can be created/canceled without immediate on-chain settlement, so metrics like "volume" should be interpreted carefully (there is room for noise).

Polymarket also explains order book concepts (bids/asks, market vs. limit orders, and spreads) in its "Using the Order Book" documentation [18].

### 2.3 Resolution: why resolution definitions are the "heart" of the market

Unlike surveys, markets require a verifiable outcome definition. Polymarket explains that markets are resolved according to written resolution rules; many markets use the UMA Optimistic Oracle as a resolution/dispute mechanism [20, 19, 16].

The practical implication is: if a question is ambiguous, the price may model the *probability of being resolved Yes* rather than the probability of the underlying substantive event.

### 2.4 Implications for using Polymarket as a "survey"

If Polymarket is used as a "survey," the "data" we should extract are not just price, but:

- price (e.g., last trade / mid price),
- volume and activity,
- spread and order book depth,
- and most importantly: the resolution rules.

## 3 Why Polymarket Can Be More Accurate Than Surveys (Theory + Evidence)

### 3.1 Incentive theory: skin in the game

When someone stakes personal money, the cost of being wrong becomes real. This encourages:

- **Higher effort:** searching for data, reading news, and comparing sources.
- **Less random answering:** careless predictions risk real losses.
- **Information aggregation:** prices summarize beliefs across many participants.

### 3.2 Empirical evidence: prediction market performance

The literature on prediction market accuracy is broad. Wolfers and Zitzewitz survey many contexts and conclude that markets often produce informative and reasonably accurate forecasts [1].

For U.S. Elections, studies of the Iowa Electronic Markets (IEM) show that market forecasts can outperform polling at longer horizons. Berg, Nelson, and Rietz (2008) compare IEM forecasts with 964 polls for U.S. Presidential elections from 1988–2004 and report that market forecasts are closer to the final outcome about 74% of the time; they also emphasize that markets outperform polls in every election when the forecast horizon is more than 100 days [4].

#### 3.2.1 Key quantitative facts (from cited sources)

To make this paper easy to read as a data-driven study, here are several key quantitative facts (all from already-cited references):

- Market vs. Poll accuracy: closer to the final outcome about 74% of the time; stronger when horizon >100 days [4].
- Polymarket on-chain dataset scale: >15,000 addresses, 4,500 events, 8,500 markets [21].

- Estimated realized arbitrage profit: about $40 million [22].
- "Semantic trading": 60–70% inter-market relation accuracy and about 20% *average returns* (˜1-week horizon) [23].
- Estimated volume distortion: about 25% of historical volume flagged as potentially fake; a peak near 60% of weekly trades in December 2024 [24].

### 3.3 Why markets can "beat" polls in some cases

Polling targets representativeness of a population, but it can fail due to nonresponse, sampling error, and modeling bias. Prediction markets target *outcomes*, not opinions. If outcomes are influenced by dispersed information (e.g., fundraising, momentum, turnout probabilities), profit-motivated traders may incorporate those signals faster.

## 4 How to Use Polymarket as a "Strict Survey" (Practical Methodology)

### 4.1 General principle: do not use a single number

A common mistake is quoting one price screenshot. To make the use more "scientific," the recommendations are:

1. collect price data regularly (e.g., hourly),
2. use statistical summaries (median/weighted mean),
3. record liquidity metrics, and
4. document major news events.

#### 4.1.1 Data collection workflow (minimal version)

To make the approach replicable, a minimal workflow consistent with Polymarket mechanics is:

1. choose a market and **copy its rules/resolution criteria** (this defines what "Yes" means) [20];
2. record the **mid price** (or last trade) regularly as an estimate of market-implied probability [18];
3. store the **spread** and (if available) bid/ask depth as a measure of signal quality [18];
4. log large moves and match them to news; if needed, check whether the move occurred in a thin-liquidity market.

### 4.2 Examples of "data" that can be extracted from public sources (and how to use them)

This section lists several numbers/cases that can be used as *evidence* in research (not to prove "Polymarket is always right," but to clarify the scale and types of risks).

#### 4.2.1 Public adoption scale

Vox summarizes the 2024 “election betting” phenomenon and describes election betting markets reaching billions of dollars (stating more than $2.3 billion) [9]. For research, such figures provide context: when many people treat market prices like “polls,” misinterpretations can have broader impact.

#### 4.2.2 Regulation as structural context

The CFTC issued a press release (Jan 3, 2022) stating that an operator of event-based binary options markets was ordered to pay a $1.4 million penalty [15]. This is relevant because regulation affects who can participate (which in turn affects representativeness and liquidity).

#### 4.2.3 Whale cases and scale of impact (politics)

The Block reports that a “French Polymarket whale” may have earned about $79 million in profits from U.S. Election bets [11], while MarketWatch discusses how such whale betting can shift market perceptions [10]. In this paper, such reporting can be used to:

- illustrate that large capital can create *price impact*,
- motivate discussion of “propaganda vs. arbitrage.”

#### 4.2.4 Resolution dispute case (oracle risk)

CoinDesk reports a resolved-market dispute around a roughly $7 million “Ukraine deal” market (Mar 27, 2025) and conflict between the Polymarket and UMA communities [14]. This supports the point that manipulation risk is not only in trading, but also in resolution.

#### 4.2.5 Wash trading estimates (risk in volume metrics)

There are two commonly cited layers of evidence. First, CoinDesk reports Fortune’s claim that Polymarket is “rife” with wash trading (Oct 30, 2024) [13].

Second, CoinDesk reports a Columbia study that analyzed more than two years of on-chain data and estimated that nearly 25% of historical volume may be “fake.” The same article mentions a peak: nearly 60% of weekly trades in December 2024 were flagged as wash trading, and a coordinated network of about 43,000 wallets [24].

For research, the main point is not that the 25% figure is certainly correct, but that: **raw volume is not a robust metric** for reading “sentiment”; it is better to use a combination of depth, open interest, and price persistence.

### 4.3 Liquidity metrics as “data quality”

Prices in thin markets can move easily due to small orders. So, to assess signal quality:

- do not record only the last price,
- also record volume, spread, and depth,
- compare price changes with volume spikes.

### 4.4 Validation: calibration and simple backtesting

To test whether Polymarket is "more correct," validation is needed. A minimal approach:

- choose a set of markets with clear outcomes (e.g., sports or political events with strong resolution definitions),
- save daily probabilities (prices),
- measure calibration: do events priced at 70% happen about 70% of the time?

### 4.5 Representativeness limitations (why this is not polling)

Even if it can be accurate for outcomes, Polymarket is not a representative sample of the population. Therefore, a safer interpretation is:

> Polymarket is an estimate from a group of traders with specific access and interests, not an estimate of population opinion.

### 4.6 Limitations and threats to validity

#### 4.6.1 Prices are not "objective" probabilities

Even though prices are often read as probabilities, the literature emphasizes that such interpretation has assumptions and limitations (e.g., risk preferences, position limits, and market frictions) [3].

#### 4.6.2 Participant bias and regulation

Polymarket participants are not representative of the population (more crypto/finance savvy) and access is constrained by regulation. For example, the CFTC documents enforcement action against Polymarket on January 3, 2022 [15]. The implication: the user base can change over time, so observed prices should be read as signals from a specific trading community, not "public opinion."

#### 4.6.3 Data quality: volume can be distorted

Volume can be distorted by wash trading. CoinDesk summarizes two types of evidence: (i) Fortune's report citing analyses by Chaos Labs and Inca Digital indicating wash trading in some markets [13], and (ii) a Columbia study estimating that about 25% of historical volume may be inauthentic, including a peak near 60% weekly volume in December 2024 [24].

#### 4.6.4 Resolution/oracle risk in ambiguous markets

Markets with subjective/ambiguous resolution definitions are more likely to trigger disputes. CoinDesk reports a roughly $7 million market that triggered a Polymarket–UMA conflict, including details about bonds and voting [14].

## 5 Whale Manipulation: Mechanisms, Evidence, and Case Studies

### 5.1 What is a whale? Two classes of manipulation

The term *whale* refers to a large-capital participant. In the Polymarket context, "manipulation" can mean:

1. **Price manipulation** (creating misleading probability signals for observers), and
2. **Resolution/oracle manipulation** (influencing how the market is resolved).

### 5.2 Mechanism 1: price impact and short-term propaganda

In markets with shallow depth, a whale can push prices up/down by consuming the order book. If the whale's goal is to influence public opinion (not pure trading profit), it can "pay" the cost as a propaganda expense. This phenomenon—pushing odds with large bets and then being corrected by other traders—is discussed in manipulation studies of betting markets (e.g., horse racing) [6].

### 5.3 Mechanism 2: wash trading and the illusion of liquidity

In some crypto markets, *wash trading* (creating artificial activity by being both buyer and seller) can make a market look active. For using Polymarket as a "survey," this is dangerous because people may overestimate signal credibility if they focus only on "volume" and "trends."

There is also more quantitative evidence: CoinDesk reports a Columbia study estimating the share of Polymarket volume that is "likely wash trading," suggesting that this can distort perceived market sentiment [24]. The CoinDesk article is not automatically final truth (detection methods can differ), but it provides *empirical evidence* that reading Polymarket as a "survey" should prioritize more robust metrics (e.g., depth, open interest, and position persistence) rather than raw volume.

### 5.4 Mechanism 3 (on-chain): resolution disputes and oracle risk

Polymarket explains its resolution integration with the UMA Optimistic Oracle for many markets [19]. In general, an optimistic oracle relies on a proposal, challenge, dispute, and voting process and is robust if participation and dispute incentives function. However, if participation is low or certain parties have outsized influence, outcomes can be contested [16].

### 5.5 Case study A: the "Ukraine deal" resolution dispute (March 27, 2025)

CoinDesk reports a roughly $7 million market that triggered a dispute between the Polymarket and UMA communities after resolution, including allegations that large UMA holders ("UMA whales") could influence voting/resolution [14]. This is an important example that "manipulation" does not only happen through trading; it can also happen at the resolution stage.

### 5.6 Case study B: the "French whale" and political odds narratives (2024)

In 2024, media highlighted a large trader (nicknamed the "French whale") placing large positions in U.S. Political markets. Some observers considered this manipulation because the orders were large and could move prices. However, Polymarket stated that an internal investigation found no evidence of manipulation; other analysis argued the pattern looked consistent with a profit-driven strategy (not simply pushing the price) [10, 12].

### 5.7 Why markets can still be robust to manipulation

Experiments by Oprea et al. (2008) show that even when traders attempt to manipulate prices to mislead observers, observers do not become systematically less accurate; markets and observers tend to be reasonably robust [5].

## 6 Conclusion and Recommendations

### 6.1 Short answer

Polymarket can serve as a more stringent "survey" signal because *skin in the game* changes incentives: participants do not merely state opinions; they bear financial consequences, which motivates accuracy. However, prices are not "pure truth" due to participant bias, liquidity, resolution design, and whale/manipulation risks.

### 6.2 Recommendations for use

1. Use Polymarket as an **additional signal**, not as a single replacement for polling.
2. Report **liquidity and resolution rules**, not only a single price.
3. Treat sharp moves in thin markets with caution; check whether supporting news exists.
4. For on-chain markets, understand that manipulation risk can shift from **trading** to **resolution**.

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